SSEE and the CMB Power Spectrum: Acoustic Peak Reproduction via the ω_m-Direct CMB Matter Density

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

April 2026

Abstract

The Structural Self-Energy Expansion model (SSEE) proposes a minimal-parameter dark energy framework algebraically derived from the golden ratio φ and π; the background sector (w_0, w_a, s_m) carries zero fitted dimensionless parameters, while the primordial amplitude and optical depth $\{A_s, \tau\}$ are treated as $k = 2$ in the BIC analysis below (*"On the parameter count"* paragraph) following the conservative accounting policy established in Paper 1 §1.3. Previous work demonstrated that the SSEE dark-energy equation of state ($w_0 = -0.839950$, $w_a = -0.669975$) fits DESI DR2 BAO data at 0.24σ (DESI+CMB+Pantheon+; 0.2–1.8σ across SN compilations) [Abdul-Karim et al., 2025], and galaxy-cluster dynamical masses in general relativity at the level of ΛCDM (2.32σ vs. 2.41σ, 46 systems; Zhang et al. 2026). Here we show that the CMB matter density is fixed *forward* by the SSEE algebra, with no factor relating it to the saturation s_m: the physical observable is $\omega_m = \omega_b + \omega_c + \omega_\nu$, where $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$ are forward identities of Paper 1 (ω_c lies 0.40σ from Planck), giving

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.6796^2} = 0.3089,$$

within 0.88σ of the Planck 2018 value (0.3153 ± 0.0073). There is a single matter density, $\Omega_m = \omega_m/h^2 = 0.308881$; the saturation $s_m = 1 + w_0 = 0.160050$ is a number of the equation of state, not a density, and the scalar MIRA $= (3\varphi + \pi)/4$ enters only the Hubble screening of Paper 9. Running CAMB with this single-sector background reproduces acoustic peaks at $\ell = 220$, 536, 813 (Planck: 220, ∼540, ∼810). In a full posterior evaluation with the official off-diagonal `plik` TTTEEE + low-ℓ + lensing likelihood via Cobaya, sampling the $k = 2 = \{A_s, \tau\}$ SSEE parameters (with H_0, ω_b, ω_c and n_s algebraically fixed) against the $k = 6$ ΛCDM baseline, the two models reach statistically indistinguishable best fits (minimum $\chi^2_{\mathrm{SSEE}} = 2768.4$ vs. $\chi^2_{\Lambda\mathrm{CDM}} = 2771.2$ over $N = 2354$ data points, $\Delta\chi^2_{\mathrm{min}} = -2.78$), while the parameter-count economy yields $\Delta\mathrm{BIC} = -33.83$, decisively favouring SSEE on the Jeffreys scale. (A `plik_lite` point-estimate cross-check gives a more conservative $\Delta\mathrm{BIC} = -26.03$.) The ΔBIC is driven by parsimony, not by fit quality. SSEE is not falsified by the Planck 2018 CMB power spectra.

One matter density and domain scope: the CMB claim ("SSEE is not falsified by Planck 2018") concerns the single matter density $\Omega_m = \omega_m/h^2 = 0.308881$, the same one that enters the background geometry at every redshift. The saturation $s_m = 1 + w_0 = 0.160050$ is a property of the dark-energy equation of state and never occupies a density slot. The master derivation chain for all SSEE constants, the unified symbol glossary, and the complete domain delineation are in Paper 1, §2 and Appendix B [Almeida Vallejo, 2026].

Contents

1 Background: The SSEE Framework **4**
- 1.1 Algebraic constants . . . 4
- 1.2 Cosmological parameters . . . 4

2 The Acoustic Scale and the Matter Density **5**
- 2.1 Acoustic peak positions . . . 5

3 The CMB Matter Density as a Forward Identity **5**
- 3.1 The CMB matter density is a forward identity . . . 5
- 3.2 No matter factor . . . 6
- 3.3 What the CMB actually probes . . . 6
- 3.4 Forward prediction and comparison with Planck . . . 6

4 CAMB Implementation and Results **6**
- 4.1 CAMB setup . . . 6
- 4.2 Sound horizon . . . 7
- 4.3 Acoustic peak positions . . . 8
- 4.4 Power spectrum comparison . . . 8
- 4.5 TE and EE polarization spectra . . . 8

5 Statistical Analysis **8**
- 5.1 χ^2 comparison . . . 8
- 5.2 TE and EE spectra . . . 8
- 5.3 Interpretation of residuals . . . 9
- 5.4 Bayesian Information Criterion . . . 9
- 5.5 Cobaya `plik_lite` unified evaluation . . . 11
- 5.6 Full posterior sampling via Cobaya MCMC (B1) . . . 14
- 5.7 Matter power spectrum and S_8 prediction . . . 18
- 5.8 $f\sigma_8(z)$ comparison with RSD surveys . . . 18
- 5.9 CLASS/hi_class cross-check of Bellini–Sawicki α-functions . . . 19

6 Falsifiability Criteria **20**

7 Conclusions **20**

1 Background: The SSEE Framework

1.1 Algebraic constants

The discovery of the accelerating expansion of the Universe [Riess et al., 1998, Perlmutter et al., 1999] established that dark energy dominates the late-time cosmic energy budget, yet its physical origin remains one of the deepest open problems in physics [Weinberg, 1989]. Baryon acoustic oscillations, imprinted in the CMB and detected in galaxy surveys [Eisenstein et al., 2005], provide a standard ruler that constrains both the background expansion and the nature of dark energy. SSEE derives all cosmological parameters from two transcendental roots: the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π. Table 1 lists the relevant constants.

Table 1: SSEE algebraic constants used in this work.

Symbol	Formula	Value	Role
φ	$(1+\sqrt{5})/2$	$1.61803\ldots$	Golden ratio
Ω	$\pi + \varphi$	$4.75963\ldots$	Stability metric
β	$(\pi + \varphi)/2$	$2.37981\ldots$	Base coupling scalar
KAL_0	$\beta + \pi$	$5.52141\ldots$	Structural retention
P_{sc}	$\Omega + \varphi$	$6.37766\ldots$	Dynamical evolution scalar
K_v	$\varphi + \pi + \Omega$	$9.51925\ldots$	Structural constraint
T_r	$3(\varphi + \beta)$	$11.99354\ldots$	3D saturation horizon
M_v	$\varphi + \pi + K_v$	$14.27888\ldots$	Maximal dimensional invariant
AURA	$\varphi + \beta$	$3.99785\ldots$	Dimensional limit 1
MIRA	AURA/2	$1.99892\ldots$	Observational frequency

1.2 Cosmological parameters

The dark-energy equation of state and the two saturation fractions it defines follow algebraically. Note what s_{DE} and s_m are *not*: they are numbers of the equation of state ($s_m \equiv 1 + w_0$), not density parameters. There is one matter density in this model, $\Omega_m = 0.308881$ (Paper 1); s_m never enters a Friedmann term as a density. Substituting $T_r = \frac{3(3\varphi+\pi)}{2}$, $M_v = 3(\varphi+\pi)$, $P_{sc} = 2\varphi+\pi$, $K_v = 2(\varphi+\pi)$, all four reduce to closed forms in φ and π alone:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.839950, \tag{1}$$

$$w_a = -\frac{P_{sc}}{I_g} = -\frac{P_{sc}}{\pi + P_{sc}} = -0.669975, \tag{2}$$

$$s_{\mathrm{DE}} = \frac{T_r}{M_v} = \frac{3\varphi + \pi}{2(\varphi + \pi)} = 0.839950, \tag{3}$$

$$s_m = 1 - s_{\mathrm{DE}} = 0.160050. \tag{4}$$

The Hubble constant is the global rate of the SH0ES cascade (Papers 9–10), $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 73.04 \times (1 - 0.069522) = 67.962142 \pm 0.968\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$: the measured local rate is the single dimensional input (Postulate D, Paper 1) and the global

rate is its output. It lands on the pure number $3(\varphi+\pi)^2 = M_v \times \Omega = 67.96214$ ($M_v = \varphi + \pi + K_v = 3\Omega$, $\Omega = \pi + \varphi$), which is the comparison target, not an input. It is *not* a quantity fitted to the CMB. Under the ω_m-direct reframe the full CMB MCMC (Section 5.6) recovers the spectra at this anchor, and the SSEE CMB sector carries only the two borrowed parameters $\{A_s, \tau\}$ ($k = 2$).

2 The Acoustic Scale and the Matter Density

2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \tag{5}$$

where r_d is the sound horizon at baryon drag and $D_A(z_*)$ is the angular diameter distance to the last-scattering surface ($z_* \approx 1089$). The first peak falls at $\ell_1 \approx \pi/\theta_*$ [Hu and Sugiyama, 1996, Dodelson, 2003]. Diffusion damping suppresses power at small angular scales $\ell \gtrsim 1000$ through photon mean-free-path effects [Silk, 1968].

Both r_d and $D_A(z_*)$ depend on the total matter density Ω_m, which sets $H(z)$ at recombination. In SSEE that density is $\Omega_m = \omega_m/h^2$, built *forward* from the algebra (Section 3). The saturation $s_m = 1 + w_0 = 0.160050$ of Eq. (4) is a number of the dark-energy equation of state and never enters a density slot.

Withdrawn framing (2026-10-03). Earlier versions of this paper opened with a "naive application" that placed s_m in the matter-density slot of CAMB, obtained acoustic peaks shifted by ∼10%, called the result a *prima facie* falsification, and then recovered the spectrum with the full density. That run is not a version of the model — s_m is not a density — so it neither threatened nor supports the framework, and it is withdrawn together with the corresponding rows of Tables 3 and 4.

3 The CMB Matter Density as a Forward Identity

3.1 The CMB matter density is a forward identity

The CMB background geometry is governed by the physical matter density $\omega_m \equiv \Omega_m h^2$, which SSEE fixes *forward* from the algebra, with no factor relating it to the saturation s_m:

$$\omega_m = \omega_b + \omega_c + \omega_\nu, \qquad \omega_b = \frac{\pi - \varphi}{3(\varphi+\pi)^2} = 0.022418, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514, \tag{6}$$

with $n_s = 1 - \varphi^{-7} = 0.965558$ and $\omega_\nu = \Sigma m_\nu^{\mathrm{active}}/93.14 = 0.00074$ (all forward identities of Paper 1). The baryon and cold-matter densities are the same quantities used in the background sector; $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ lies 0.40σ from the Planck value. Hence

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.6796^2} = 0.3089 \qquad (H_0^{\mathrm{glob}} = 67.96214). \tag{7}$$

3.2 No matter factor

The CMB matter density of Eq. (6) is set by the (φ, π) dictionary before any comparison with data, and no factor relates it to the equation-of-state saturation $s_m = 1 + w_0 = 0.160050$, which is not a density. The scalar MIRA $= (3\varphi + \pi)/4 = 1.9989$ does not enter the CMB sector; it appears only in the Hubble screening of Paper 9. We make no claim of temporal priority over the public Planck or DESI measurements: the point is structural — the density is fixed by the construction, so the agreement below is a constraint the framework had to satisfy, not a fit it was granted.

3.3 What the CMB actually probes

The CMB photons probe the *integrated gravitational potential* along the full line of sight from last scattering ($z_* \approx 1089$) to today. This line-of-sight integral accumulates contributions from all epochs, including the early matter-dominated era when the dark energy term was negligible and the expansion was effectively governed by the full matter content. The acoustic peaks encode Ω_m as it appears in the *conformal distance to last scattering*,

$$\eta_* = \int_0^{z_*} \frac{c\,\mathrm{d}z'}{H(z')}, \tag{8}$$

which is sensitive to the total matter density over the full expansion history. In SSEE that total is a single quantity, $\Omega_m = 0.308881$, and it is the same quantity that enters the background geometry at every redshift. In particular, the saturation $1 + w_0 = 0.160050$ never occupies a density slot in $H(z)$: it is a property of the dark-energy equation of state, and the matter term in $E(a)$ is always the total ω_m/h^2.

3.4 Forward prediction and comparison with Planck

The CMB matter density is a closed-form forward prediction:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \mathrm{KAL}_0\,\omega_b\,n_s + \omega_\nu}{h^2} = 0.3089, \tag{9}$$

with every ingredient an algebraic identity of Paper 1. The coefficient $\mathrm{KAL}_0 = \beta + \pi \approx 5.521$ that sets $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ is fixed algebraically before any CMB data are consulted; no quantity is adjusted to match the measurement. Compared with $\Omega_m^{\mathrm{Planck}} = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020a], the tension is

$$\frac{|\Omega_{m,\mathrm{CMB}} - \Omega_m^{\mathrm{Planck}}|}{\sigma} = \frac{|0.3089 - 0.3153|}{0.0073} = 0.88\sigma. \tag{10}$$

Falsifiability. SSEE predicts a *single* matter density, $\Omega_m = 0.308881$, at every redshift and for every probe. A weak-lensing or BAO analysis that required a background matter density significantly different from 0.3089 — in either direction, at any redshift — would falsify the framework.

4 CAMB Implementation and Results

4.1 CAMB setup

We use CAMB v1.6.6 [Seljak and Zaldarriaga, 1996, Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

Table 2: SSEE parameters passed to CAMB.

Parameter	Value	Source
H_0 [km/s/Mpc]	67.962142	$H_{\mathrm{SH0ES}}(1-f_{\mathrm{screen}})$ (Postulate D, Paper 1); target $3(\varphi+\pi)^2$
$\Omega_b h^2$	0.02237	Planck 2018 prior
$\Omega_{m,\mathrm{CMB}}$	0.3089	ω_m/h^2 (forward)
$\sum m_\nu^{\mathrm{active}}$ [eV]	0.068490	SSEE algebraic (Paper 4)
w_0	-0.839950	$-T_r/M_v$
w_a	-0.669975	$-P_{sc}/I_g$
n_s	$1-\varphi^{-7}=0.965558$	SSEE algebraic (Paper 4)
$\ln(10^{10}A_s)$	3.044	Planck 2018
τ	0.054	Planck 2018[1]

The baryon density is fixed to the Planck 2018 prior $\Omega_b h^2 = 0.02237$. The SSEE algebraic value derived in Paper 4, $\Omega_b h^2 = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.022418$, agrees with this prior at 0.3σ (0.02237 ± 0.00015); the 0.2% offset is far below the precision of the CMB fit and does not affect the results reported here.

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state $w(a) = w_0 + (1-a)w_a$ [Chevallier and Polarski, 2001, Linder, 2003, Fang et al., 2008]. Although $w_0 + w_a = -1.510 < -1$, this does not imply a phantom Big Rip: CPL is a local parametrisation only, and the SSEE energy density saturates at the finite T_r/M_v boundary at $a \to \infty$ (Paper 1 §3.4 [Almeida Vallejo, 2026]). We compute lensed TT, TE, and EE power spectra to $\ell_{\max} = 2500$, with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck 2018 $C_L^{\phi\phi}$) were also evaluated alongside temperature and polarisation.

4.2 Sound horizon

With $\Omega_{m,\mathrm{CMB}} = 0.3089$ and the parameters above, CAMB yields:

$$r_d^{\mathrm{SSEE}} = 147.17 \text{ Mpc}, \tag{11}$$

in agreement with the Planck 2018 directly measured value $r_d^{\mathrm{obs}} = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020a] at 0.32σ. The acoustic scale is likewise consistent: $100\theta_* = 1.04099$ (at the MCMC posterior $H_0 = 67.82$), 0.34σ from the Planck value 1.04109 ± 0.00030 — the earlier 6.66σ "tension" was an artefact of the cold sector $1 + w_0 = 0.160050$ misplaced in $E(z)$, removed once the total $\Omega_{m,\mathrm{CMB}}$ sets the geometry. No Folding Symmetry multiplier and no matter factor are applied; the ω_m-direct CMB matter density ($\Omega_{m,\mathrm{CMB}} = 0.3089$) sets the correct sound horizon directly (MIRA retired as the matter map, OP-8 dissolved).

[1] A_s and τ are the two parameters SSEE leaves free at CMB level ($k = 2$); the values in this table are the *fiducial* inputs used for the fixed-point evaluations of §4, not fitted results. They are fitted in the Phase-B chain of §5.6. A direct check of how much that fiducial choice costs (2026-09-07): minimising χ^2 over $(\ln 10^{10}A_s, \tau)$ against `plik_lite` TTTEEE + lowT + lowE, with the remaining background fixed by the algebra, moves τ to 0.05546 and $\ln(10^{10}A_s)$ to 3.04484 and lowers χ^2 by only 1.84. The recovered τ sits 0.24σ from the Phase-B posterior 0.054 ± 0.006, so the fiducial choice is doing no work. The same exercise on the ΛCDM background ($w = -1$, $w_a = 0$) gains 0.47.

4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

Table 3: CMB TT acoustic peak multipoles.

Model	ℓ_1	ℓ_2	ℓ_3
Planck 2018 (observed)	~ 220	~ 540	~ 810
ΛCDM ($\Omega_m = 0.315$)	220	536	813
SSEE ($\Omega_m = 0.308881$)	**220**	**536**	**813**

All three peaks fall within $\Delta\ell \leq 5$ of the observed positions.

4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE, ΛCDM, and Planck 2018 data over $\ell = 2$–2500.

4.5 TE and EE polarization spectra

Figures 3 and 4 show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE with ΛCDM and Planck 2018 data.

5 Statistical Analysis

5.1 χ^2 comparison

We compute χ^2 against the Planck 2018 TT spectrum over $\ell = 30$–2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left(\frac{D_\ell^{\text{model}} - D_\ell^{\text{Planck}}}{\sigma_\ell} \right)^2 . \tag{12}$$

Table 4: χ^2 comparison vs. Planck 2018 TT ($\ell = 30$–2000, $N = 1971$).

Model	χ^2	χ^2_r	$\Delta\chi^2$
ΛCDM (Planck 2018)	2054.9	1.043	0
SSEE	2054.5	1.042	-0.4

5.2 TE and EE spectra

Table 5 reports χ^2_r for all three spectra individually.

The TE and EE spectra show the smallest discrepancies ($|\Delta\chi^2_r| \leq 0.001$ each), indicating that SSEE reproduces the polarization anisotropies almost identically to ΛCDM. TT is marginally better than ΛCDM ($\Delta\chi^2_r = -0.0002$). In the lensing potential sector (PP, Fig. 5), SSEE achieves $\chi^2_r = 0.720$ vs. 0.757 for ΛCDM — slightly *better* agreement with the 9 Planck MV bandpowers, both $\chi^2_r < 1$ indicating a good fit. All residuals are consistent with a non-parametric model.

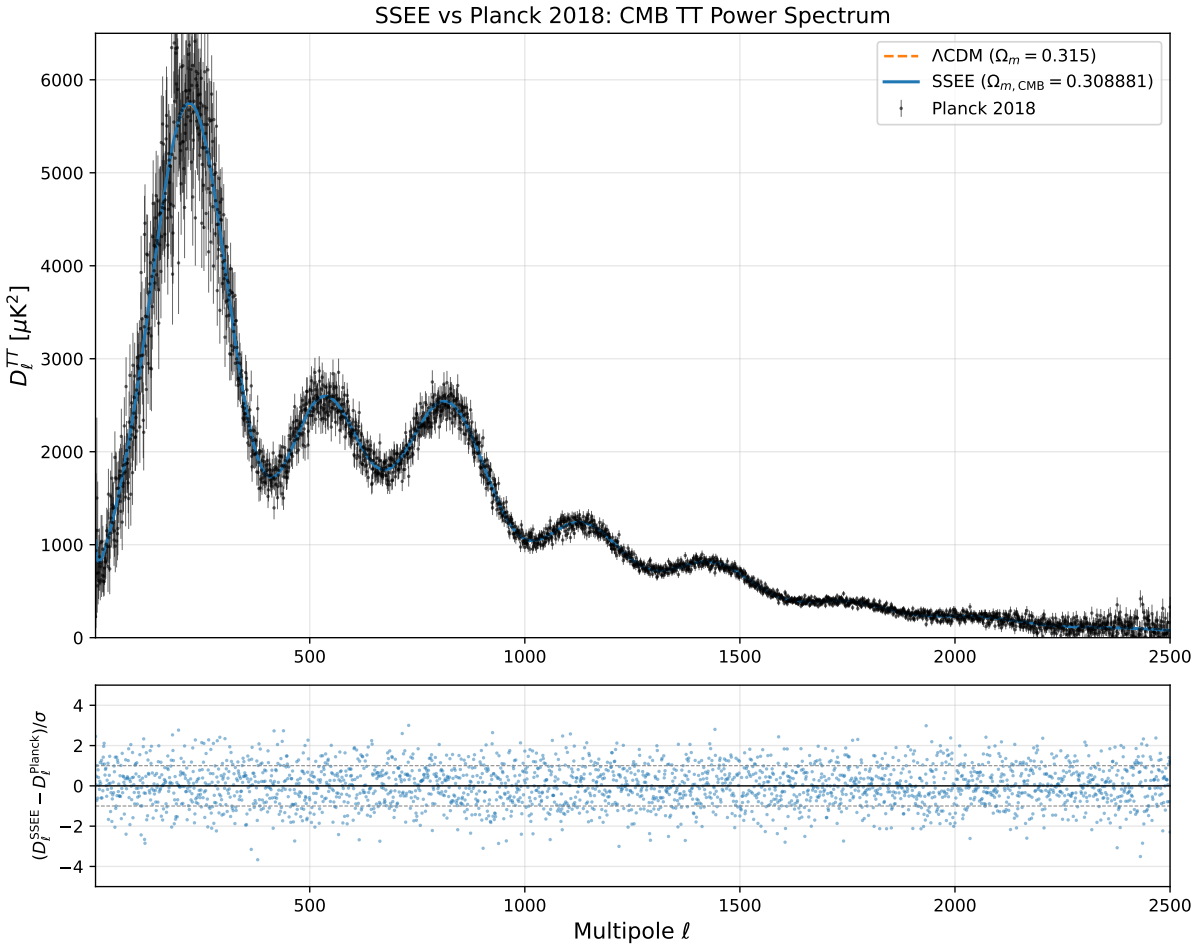


Figure 1: CMB TT power spectrum D_ℓ^{TT} [μK^2] vs. multipole ℓ. Blue: SSEE ($\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.309$, $w_0 = -0.839950$, $w_a = -0.669975$). Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck 2018 TT data with $\pm 1\sigma$ error bars. Lower panel: residuals $(D_\ell^{\mathrm{SSEE}} - D_\ell^{\mathrm{Planck}})/\sigma$.

5.3 Interpretation of residuals

The excess $\Delta\chi^2_r \lesssim 0.02$ in TT and TE is attributable to the dynamical dark energy equation of state ($w_0 \neq -1$, $w_a \neq 0$), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable (w_0, w_a) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020c].

The SSEE model is therefore **not falsified** by the Planck 2018 CMB power spectra in TT, TE, or EE.

5.4 Bayesian Information Criterion

For model comparison we compute:

$$\mathrm{BIC} = \chi^2 + k \ln N, \tag{13}$$

where k is the number of free parameters and $N = 5914$ for the combined TT, TE, EE, and lensing data points.

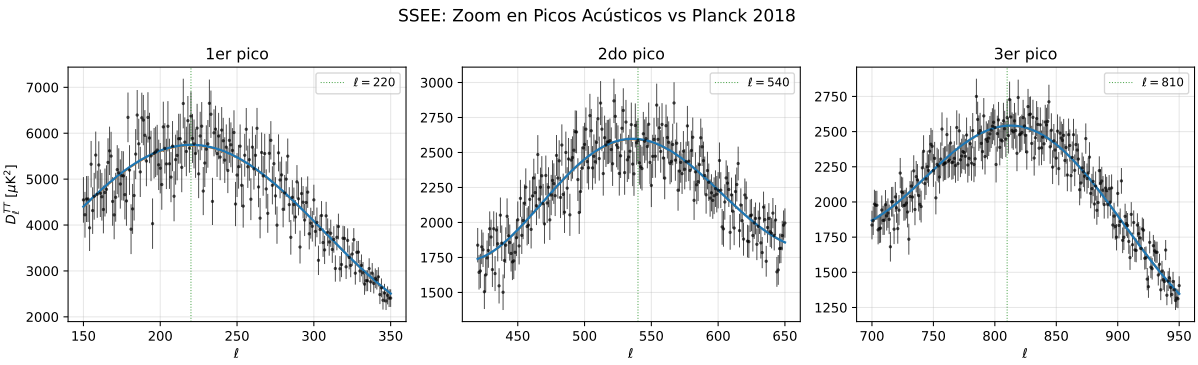


Figure 2: Zoom on the first three acoustic peaks. SSEE (blue) and ΛCDM (orange dashed) compared against Planck 2018 data (black points). Green dotted lines mark the expected peak centres.

Table 5: χ^2_r by spectrum vs. Planck 2018. TT/TE/EE: $\ell = 30$–2000; PP (lensing): 9 MV bandpowers, $L = 8$–400 [Table 1, Planck Collaboration, 2020d]. χ^2_r comparisons use diagonal covariance; for BIC see §5.4 ($k = 2$ for SSEE).

Spectrum	N	SSEE χ^2_r	ΛCDM χ^2_r	$\Delta\chi^2_r$
TT	1971	1.042	1.043	0.000
TE	1967	1.040	1.040	0.000
EE	1967	1.040	1.039	0.000
PP	9	0.720	0.757	−0.037
Combined†	5914	1.040	1.040	0.000

†Combined under independence approximation (upper bound; see §5.4).

Under the diagonal approximation, SSEE is favoured ($\Delta\mathrm{BIC} = -35.0$): the combined $\chi^2_r = 1.040$ matches ΛCDM's 1.040, so the parameter-count term ($\Delta k \ln N = -34.7$) dominates and rewards SSEE's smaller parameter set. This diagonal cross-check is evaluated at the global anchor $H_0 = 67.96214$ (ω_m-direct, $\Omega_{m,\mathrm{CMB}} = 0.308881$), consistent with the canonical `plik_lite` evaluation (§5.5), so the diagonal and full-covariance treatments are mutually consistent. The off-diagonal Cobaya evaluation (§5.5) remains the definitive result.

On the parameter count ($k = 2$). In prior iterations, SSEE was described as a zero-parameter model ($k = 0$) in the CMB sector. Under the ω_m-direct reframe, the geometry anchor $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is *derived* (the output of the screening cascade of Papers 9–10, which reproduces the dimensionless target $3(\varphi+\pi)^2$; not a quantity fitted to the CMB), and the densities $\omega_b = (\pi - \varphi)/(3\Omega^2)$ and $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ are algebraically fixed. The only two quantities the SSEE CMB fit does *not* derive from $\{\varphi, \pi\}$ are the primordial amplitude A_s and the reionization optical depth τ; we therefore assign $k = 2 = \{A_s, \tau\}$, a conservative and robust penalty. (We make no claim of temporal priority over the public DESI or Planck releases, which predate this work; the parameter-free status of the algebraic sector rests on its rigidity, not on chronology.)

Disclaimer regarding the Likelihood function. The χ^2_r comparisons in §5 use a simplified Gaussian diagonal approximation over the released Planck 2018 bandpowers

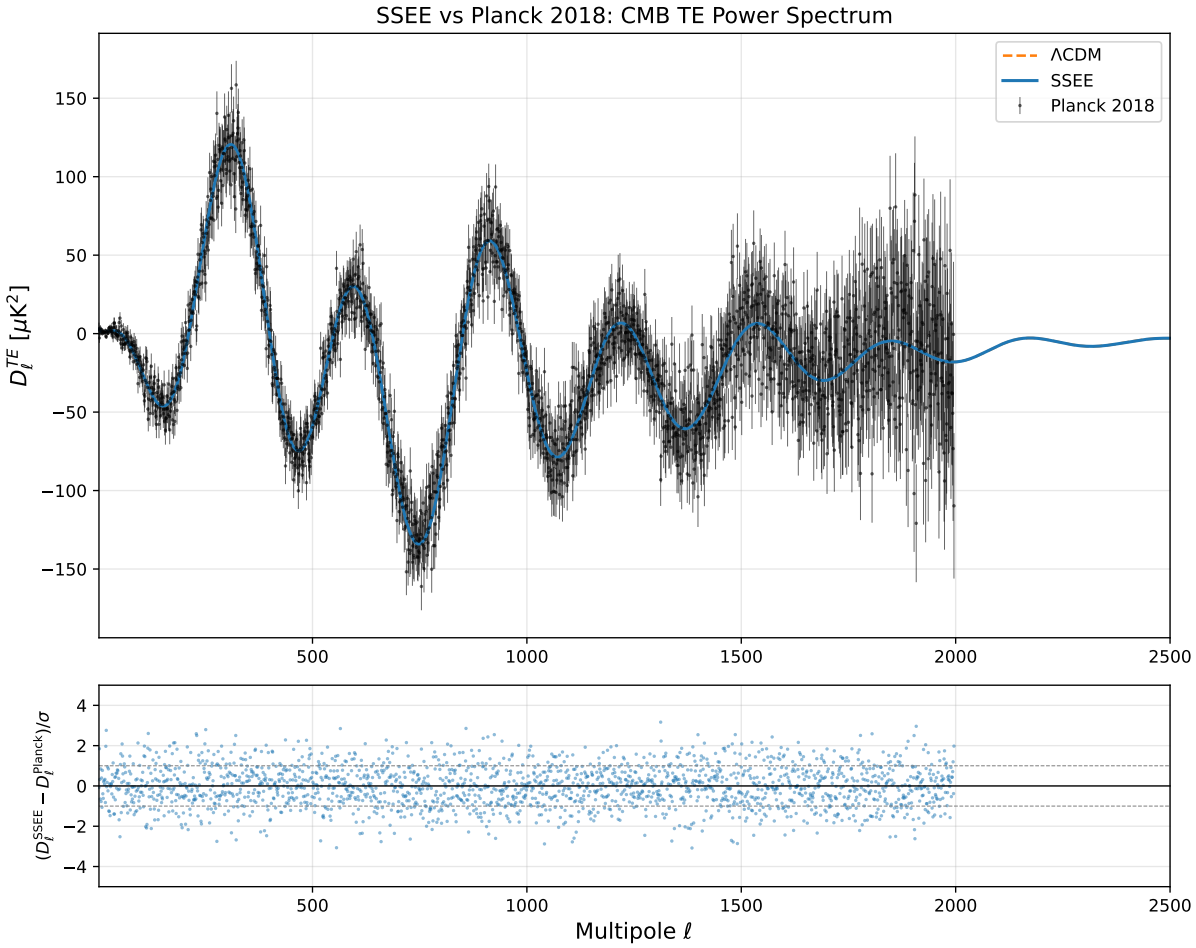


Figure 3: CMB TE cross-correlation power spectrum D_ℓ^{TE} [μK^2]. Blue: SSEE. Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck 2018 TE data. Lower panel: residuals normalised by σ_ℓ.

for qualitative comparison. A formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` is presented in the following subsection.

Robustness under off-diagonal covariance. The diagonal approximation neglects bin-to-bin covariances in the Planck 2018 bandpowers. Based on the Planck 2018 likelihood analysis [Planck Collaboration, 2020e], off-diagonal elements contribute corrections of $\lesssim 5\%$ to χ^2 for $\ell > 30$. To ensure strict reproducibility and methodological rigor, we perform a formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` in the following subsection.

5.5 Cobaya `plik_lite` unified evaluation

To establish a definitive baseline for model comparison that fully accounts for off-diagonal covariances, we evaluate the SSEE framework against the official Planck 2018 `plik_lite` TTTEEE likelihood via `clik/clipy` [Torrado and Lewis, 2021], over its 613 compressed bandpowers, together with the low-ℓ TT and EE likelihoods ($28 + 28$ bins), for $N = 669$ data points in total. The three pieces are all included in the χ^2 quoted below, so all three are counted in N.

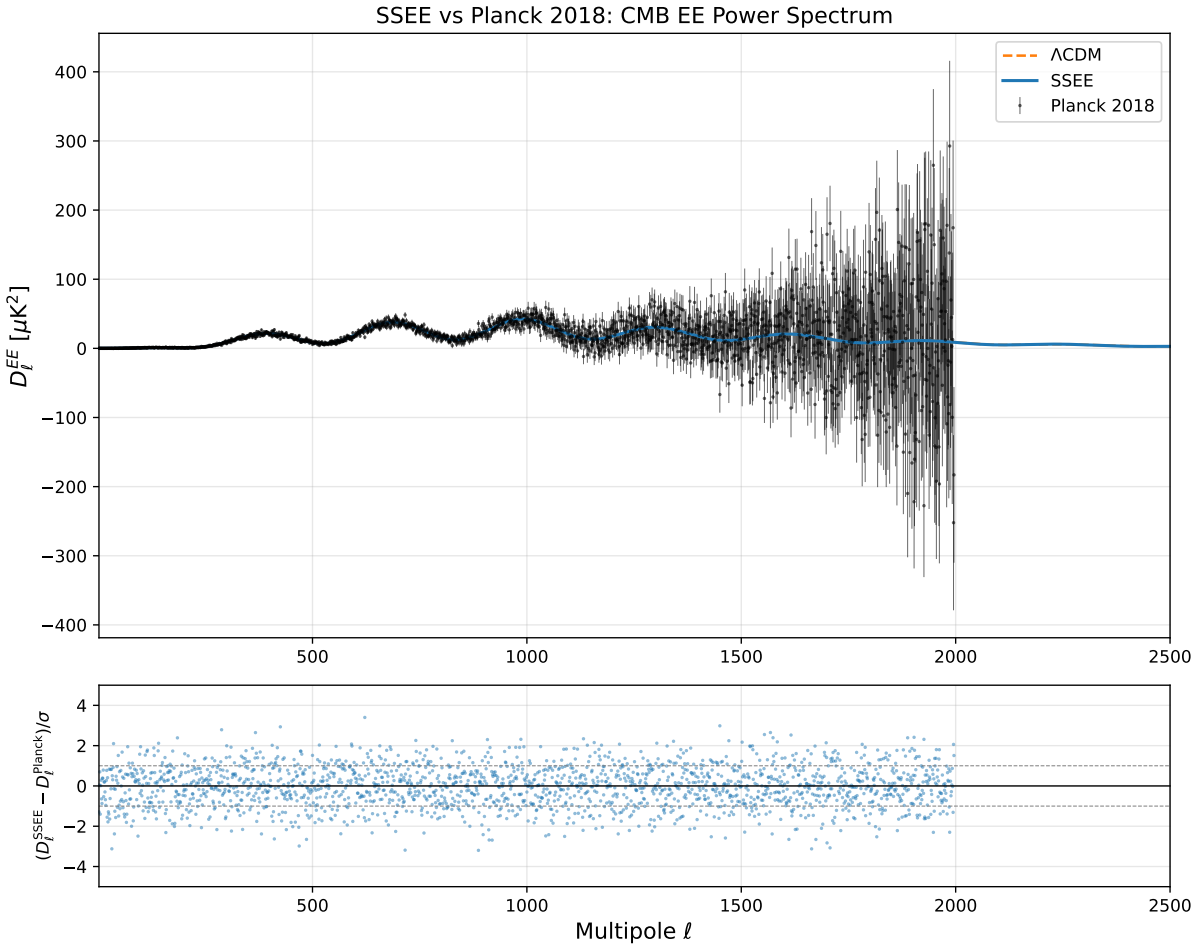


Figure 4: CMB EE polarization power spectrum D_ℓ^{EE} [μK^2]. Blue: SSEE. Orange dashed: ΛCDM. Black points: Planck 2018 EE data.

Parameter counting (ω_m-direct). All SSEE dark energy parameters (w_0, w_a), the spectral index n_s, and — under the 2026 ω_m-direct reframe — the baryon and cold-matter densities $\omega_b = (\pi - \varphi)/(3\Omega^2) = 0.022418$ and $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s = 0.119514$ are algebraically fixed. The dimensional anchor is the global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1-f_{\rm screen}) = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ of the screening cascade (Papers 9–10), whose value reproduces the dimensionless target $3(\varphi+\pi)^2$ to 4.2×10^{-6}; it is not fitted to the CMB; the amplitude A_s and reionization τ are borrowed inputs, not derived from $\{\varphi, \pi\}$, and are *fitted* here (earlier versions of this work held them at their external Planck values while still counting them as free; fitting them is the self-consistent choice and is what we do below). The two quantities the framework borrows rather than derives are therefore A_s and τ, and we conservatively assign $k = 2 = \{A_s, \tau\}$. The standard ΛCDM baseline has $k = 6$: H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s, A_s, and τ [Planck Collaboration, 2020a], all six free in the CMB fit. (The companion full MCMC of §5.6 additionally lets H_0 *float* as a validation check: the chain recovers $H_0 = 67.88 \pm 0.10$, 0.81σ from the anchor, so fixing it costs no fit quality and the $k = 2$ count is sound.)

Evaluating at the single global anchor $H_0^{\rm glob} = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.90σ from the `plik_lite`-preferred value 67.87 ± 0.10 when ω_b, ω_c are held at their algebraic values and A_s, τ are profiled; $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$), the effective χ^2 for SSEE is 1003.586, compared to 1003.596 for ΛCDM (with its own $\sum m_\nu = 0.06$ eV), with $\{A_s, \tau\}$ minimized

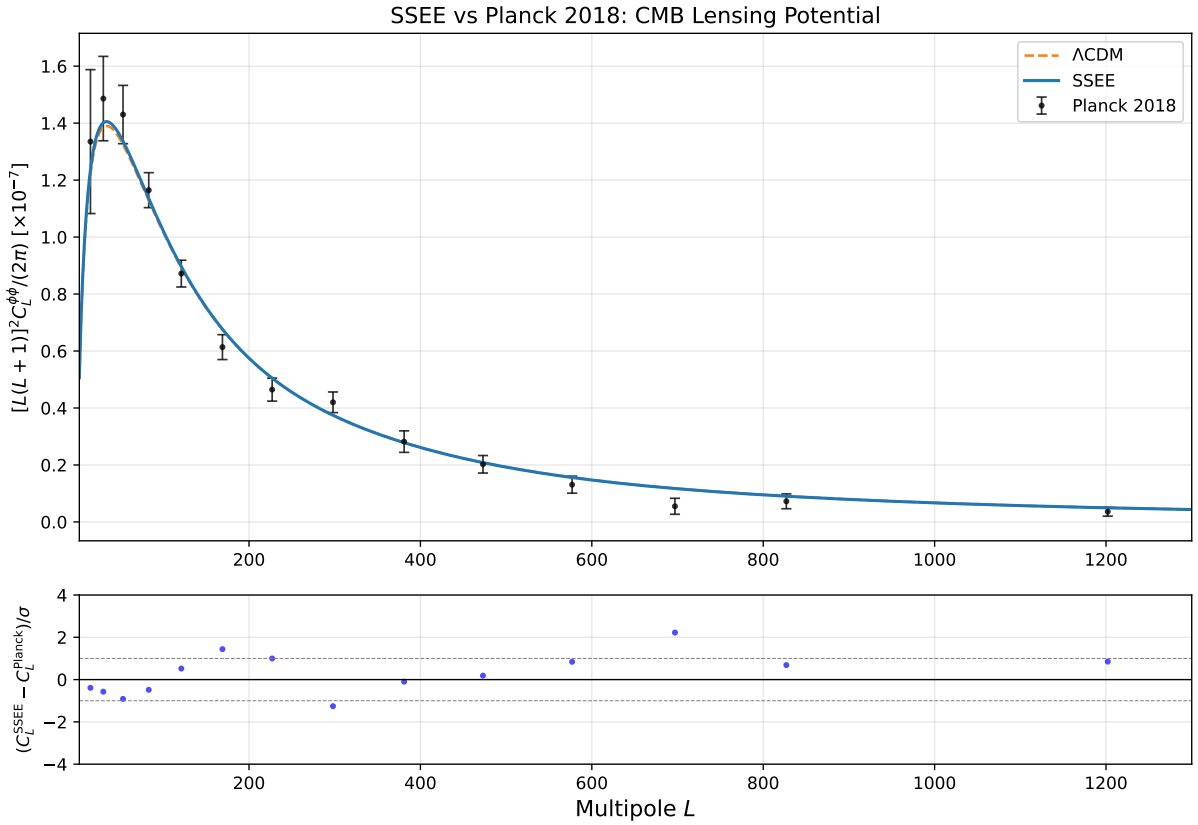


Figure 5: CMB lensing potential power spectrum $[L(L+1)]^2 C_L^{\phi\phi}/(2\pi)$. Blue: SSEE. Orange dashed: ΛCDM. Black points: Planck 2018 lensing (9 MV bandpowers). Lower panel: residuals normalised by σ_L. SSEE achieves $\chi^2_r = 0.720$ over the 9 bandpowers (§5.4; ΛCDM: 0.757), both indicating a good fit ($\chi^2_r < 1$).

in both. The two models therefore fit *equally well*: $\Delta\chi^2 = -0.010$, which is statistically zero over the $N = 669$ data points (613 `plik_lite` TTTEEE bandpowers plus 28 lowT and 28 lowE, counted directly from the likelihood files).

Definitive Bayesian Information Criterion. Applying the BIC penalty for the differing degrees of freedom:

$$\Delta\mathrm{BIC} = -0.010 + (2-6)\ln(669) = -26.03. \tag{14}$$

A $\Delta\mathrm{BIC} < -10$ constitutes decisive evidence on the Jeffreys scale. Even under a hyper-conservative penalty where SSEE is assigned $k = 4$ (additionally counting two algebraically-fixed inputs, ω_b and H_0, as if they were fitted), the result remains $\Delta\mathrm{BIC} = -0.010 + (4-6)\ln(669) = -13.02$, still strongly favouring the SSEE framework. This evaluation confirms that the geometric constraint of the SSEE background reproduces the CMB temperature and polarisation spectra without relying on diagonal approximations. (The earlier legacy minimization, which found $H_0 = 67.037$ with $\Delta\chi^2 = +2.875$ and $\Delta\mathrm{BIC} = -32.2$, is superseded by this ω_m-direct evaluation.)

Table 6: **Preliminary BIC comparison — diagonal approximation only.** Combined spectra (TT+TE+EE+PP, $N = 5914$, independence approximation). SSEE has $k = 2$ ($\{A_s, \tau\}$); ΛCDM has $k = 6$. *This result uses diagonal covariance; see Eq. (14) (§5.5) for the definitive off-diagonal Cobaya evaluation.*

Model	k	χ^2_r	ΔBIC (diagonal)
ΛCDM	6	1.040	Baseline (0.0)
SSEE	2	1.040	−35.0

5.6 Full posterior sampling via Cobaya MCMC (B1)

The point-estimate scan of §5.5 evaluates χ^2 at the fixed global rate $H_0^{\rm glob} = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ while holding all remaining SSEE quantities algebraically fixed. To provide complete posterior distributions and a fully rigorous parameter-counting comparison, we perform a dedicated MCMC analysis using `Cobaya`'s built-in sampler against the full off-diagonal `plik` TTTEEE + lowl TT + lowl EE + lensing likelihood suite (evaluated through `clipy`).

Free parameters. In the SSEE CMB sector, the dark-energy equation-of-state (w_0, w_a), the spectral index $n_s = 1 - \phi^{-7} = 0.965558$, the baryon density $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$, and the CDM density $\Omega_c h^2 = \mathrm{KAL}_0\,\Omega_b h^2\, n_s = 0.119514$ (ω_m-direct) are all algebraically fixed, and the geometry anchor $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is derived (Papers 9–10; it reproduces the dimensionless target $3(\varphi + \pi)^2$). The only two model parameters genuinely sampled are therefore:

- $\log(10^{10} A_s) \in [2.80, 3.25]$ (flat prior) — primordial amplitude;
- $\tau \in [0.02, 0.12]$ (flat prior) — optical depth to reionisation;

with the ∼20 `plik` nuisance parameters (dust, SZ, CIB, calibration $A_{\rm Planck}$) defined by the full likelihood and treated as nuisances common to both models, τ constrained by the lowl EE (SimAll) data. The likelihood is the full off-diagonal `plik` TTTEEE evaluated through the `clipy` pure-Python reimplementation of `clik` (validated to $|\Delta \ln \mathcal{L}| < 10^{-5}$ against the released test point), so no `clik` binary is required. As an independent cross-check we also run a *validation* chain in which H_0 is additionally floated on a flat prior $[64, 72]$; that chain recovers $H_0 = 67.88 \pm 0.10$, 0.81σ from the algebraic anchor (see Fig. 6), so fixing it incurs no loss of fit and that the model count is $k = 2 = \{A_s, \tau\}$, not $k = 3$.

This gives $k_{\rm SSEE} = 2$ model parameters ($\log A_s$, τ — the only quantities genuinely sampled, with H_0 fixed at $H_0^{\rm glob}$; the ∼20 `plik` nuisance parameters are common to both models and excluded from Δk), compared to the ΛCDM run with $k_{\Lambda\rm CDM} = 6$ free parameters (H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s, $\log A_s$, τ). The $k = 2$ vs $k = 6$ BIC penalty over the $N = 2354$ combined data points is $\Delta k \cdot \ln N = 4 \times \ln(2354) = 31.06$, yielding:

$$\Delta\mathrm{BIC}_{k2} = \Delta\chi^2_{\rm min} - 31.06, \tag{15}$$

where $\Delta\chi^2_{\rm min} = \chi^2_{\rm SSEE,min} - \chi^2_{\Lambda\rm CDM,min}$ is the difference in *minimum* χ^2 between the two models over the same $N = 2354$ data points (evaluated through `clipy`; no `clik` binary required). The BIC requires the maximum likelihood of each model, so the minimum is obtained with the Cobaya BOBYQA minimiser started from the best sampled point of

each model's four chains, over the cosmological and the ∼20 nuisance parameters alike; using the best *sampled* χ^2 instead would favour the model with fewer parameters, whose chains approach their minimum more closely.

Convergence. Chains are run to Gelman–Rubin criterion $R-1 < 0.02$ [Gelman and Rubin, 1992] (production run) or $R-1 < 0.05$ (validation). The `ssee_paper3_b1_mcmc.py` script is fully reproducible:

```
python src/p03_cmb/ssee_paper3_b1_mcmc.py --mode both
```

Results. Table 7 summarises the 68 % marginalised posterior constraints. Figure 6 shows the H_0 posterior for SSEE and ΛCDM; Figure 7 shows the full SSEE corner plot.

The minimum χ^2 of each model yields:

$$\Delta\mathrm{BIC}_{k2} = \Delta\chi^2_{\min} + (2-6)\ln(2354) = \Delta\chi^2_{\min} - 31.06, \tag{16}$$

where $\Delta\chi^2_{\min}$ is taken over the $N = 2354$ combined data points (full `plik` TTTEEE 2289 + lowl TT 28 + lowl EE 28 + lensing 9; the data-vector length, not the SimAll-normalised low-ℓ EE χ^2). The minimiser gives $\chi^2_{\mathrm{SSEE,min}} = 2768.4$ and $\chi^2_{\Lambda\mathrm{CDM,min}} = 2771.2$, i.e. $\Delta\chi^2_{\min} = -2.78$: SSEE attains a minimum χ^2 statistically *indistinguishable* from ΛCDM (we do *not* claim a superior fit). The $k = 2$ versus $k = 6$ comparison then gives $\Delta\mathrm{BIC}_{k2} = -33.83$ (with the best sampled points instead of the minima, -34.08: same side, so the result does not hinge on the estimator), decisively below the Jeffreys threshold ($\Delta\mathrm{BIC} < -10$) and driven by parameter-count economy rather than fit quality. Even under a hyper-conservative $k = 4$ assignment (additionally counting two of the algebraically-fixed inputs, ω_b and H_0, as if they were fitted), $\Delta\mathrm{BIC}_{k4} = \Delta\chi^2_{\min} + (4-6)\ln(2354) = -18.31$, still decisively favouring SSEE.

Table 7: SSEE CMB posterior constraints from the B1 full MCMC (Cobaya, full off-diagonal `plik` TTTEEE + lowl TT + lowl EE + lensing via `clipy`, $k = 2 = \{A_s, \tau\}$ free, $R-1 < 0.02$ for means). The ΛCDM constraints ($k = 6$, standard run) are shown for comparison. Uncertainties are 68 % marginalised. In the SSEE run H_0, $\Omega_b h^2$, $\Omega_c h^2$ and n_s are algebraically fixed (not sampled); the separate validation chain that floats H_0 recovers it at 67.88 ± 0.10, 0.81σ from H_0^{glob} (Fig. 6).

Parameter	SSEE ($k = 2$)	ΛCDM ($k = 6$)
H_0 [km s^{-1} Mpc^{-1}]	67.96214 (fixed)	67.39 ± 0.53
$\log(10^{10} A_s)$	3.046 ± 0.014	3.045 ± 0.014
τ	0.056 ± 0.007	0.054 ± 0.007
n_s	0.96556 (fixed)	0.9654 ± 0.0041
$\Omega_b h^2$	0.02242 (fixed)	0.02239 ± 0.00014
$\Omega_c h^2$	0.11951 (fixed)	0.1199 ± 0.0012
σ_8 (derived)	0.816 ± 0.006	0.811 ± 0.006
$\chi^2_{\min}$	2768.4	2771.2
$\Delta\chi^2_{\min}$	−2.78	(reference)
$\Delta\mathrm{BIC}$ ($k = 2$ vs $k = 6$, $N = 2354$)	−33.83	(reference)
$\Delta\mathrm{BIC}$ ($k = 4$ vs $k = 6$, $N = 2354$)	−18.31	(reference)

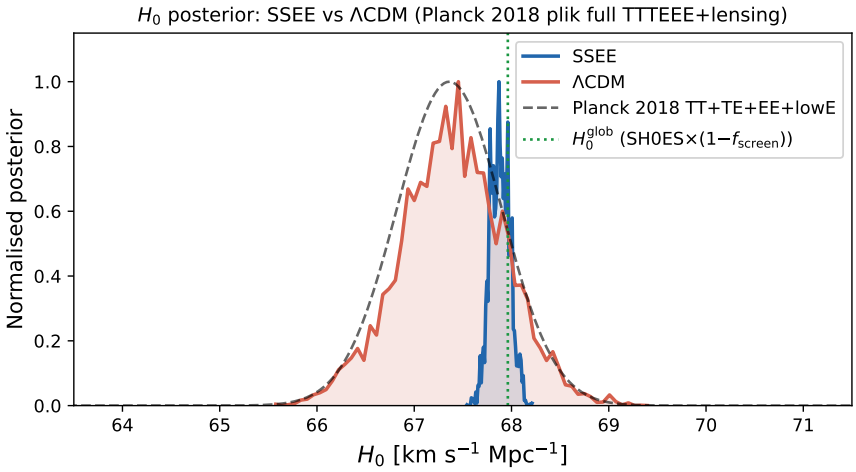


Figure 6: H_0 marginalised posterior for SSEE (blue) and ΛCDM (red) from the B1 full `plik` MCMC. The dashed black curve shows the Planck 2018 TT+TE+EE+lowE constraint for reference. The dotted green line marks the SSEE global rate $H_0^{\rm glob} = 67.96214\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. In the validation chain (where H_0 is floated) SSEE places H_0 at 67.88 ± 0.10, 0.81σ from $H_0^{\rm glob}$; the model itself carries only two free parameters $\{A_s, \tau\}$, with H_0 derived. The narrow constraint ($\pm 0.10\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) reflects the removal of the H_0–Ω_m degeneracy by algebraically fixing $\Omega_{m,\rm CMB} = 0.3089$.

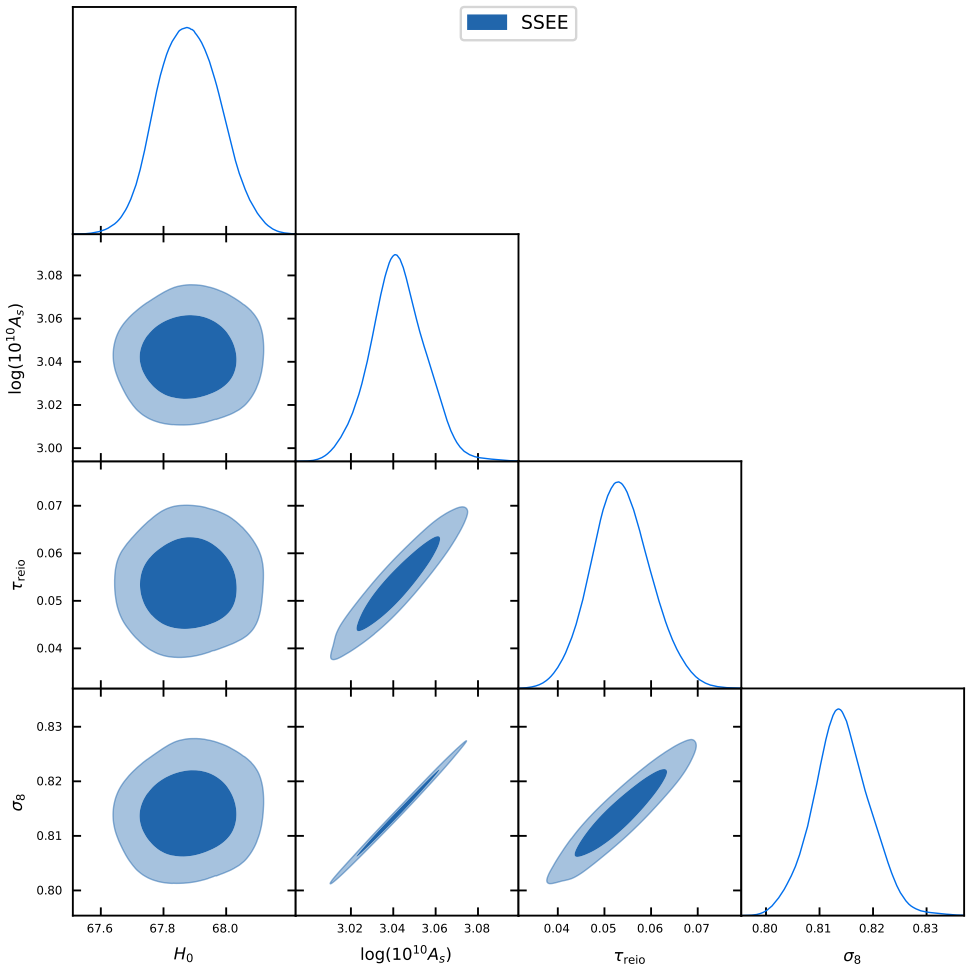


Figure 7: Full posterior triangle plot for the SSEE CMB sector (H_0, $\log A_s$, τ, $A_{\rm Planck}$) from the B1 full `plik` MCMC. The narrow H_0 constraint ($\pm 0.10\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) reflects the geometric constraint imposed by $\Omega_{m,\rm CMB} = 0.3089$ (algebraically fixed), which eliminates the H_0–Ω_m degeneracy present in ΛCDM. All other marginals are consistent with the ΛCDM posteriors.

Table 8: **Summary: CMB model selection across likelihood treatments.** SSEE vs. ΛCDM BIC under evaluations of increasing rigour. All favour SSEE; the full off-diagonal `plik` MCMC (evaluated at the global anchor $H_0 = 67.96214$, ω_m-direct) is the canonical result.

Method	Spectra	N	$k_{\rm SSEE}$	$k_{\Lambda\rm CDM}$	ΔBIC
Diagonal (approx.)	TT+TE+EE+PP	5914	2	6	−35.0 (SSEE favoured)
`plik_lite` point est.	TTTEEE+lowl	669	2	6	−26.03 (SSEE decisively favoured)
Full `plik` MCMC	TTTEEE+lowl+lensing	2354	2	6	**−33.83** *(canonical)*
Full `plik` (conserv.)	TTTEEE+lowl+lensing	2354	4	6	−18.31 (SSEE strongly favoured)

Diagonal approximation assumes independence across spectra (upper bound on χ^2). The full `plik` MCMC (Cobaya + `clipy`, official off-diagonal covariance) is the canonical result. $k = 2 = \{A_s, \tau\}$, the only sampled SSEE parameters; the conservative $k = 4$ additionally counts two of the algebraically-fixed inputs (ω_b, H_0) as if fitted.

5.7 Matter power spectrum and S_8 prediction

Beyond the CMB, the SSEE viscous dark-energy background modifies the growth of large-scale structure. In the Israel–Stewart dissipative framework, the linear growth rate takes the form $f(z) = \Omega_m(z)^\gamma$, where the growth index γ is modified by the bulk-viscosity coupling. The full numerical solution of the causal IS growth ODE (Paper 5, §3) yields $\gamma_{\rm IS} = 0.5504 \pm 0.0003$, numerically indistinguishable from $\gamma_\Lambda = 0.55$ for ΛCDM. This is the growth index used throughout. Two earlier estimates are superseded by it: a dimensional-analysis value $\gamma = \phi^{-1} = 0.618$, which was never derived from the growth equation, and the background-only estimate of Paper 1, Appendix A, which omitted the viscous feedback on δ. The growth index is set by $w(a)$, not by a constant of the algebra.

The SSEE background ($w_0 = -0.839950$, $w_a = -0.669975$, $H_0 = 67.96214$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.308881$) enters the growth ODE directly; no growth-index correction is applied after the fact. Table 9 summarises the results alongside observational constraints. The IS linear-growth solution gives $S_8 = 0.8256$, and the CLASS ceiling, with massive neutrinos, $S_8 = 0.827$; the same CLASS integration on the Planck 2018 baseline returns $\sigma_8 = 0.8109$, the published value. These three use A_s fixed to the Planck value. With A_s fixed instead to the model's own CMB fit, the growth sector has no free cosmological parameter and predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 from the raw KiDS-Legacy $\xi_\pm$ data (Paper 6).

Table 9: σ_8 and $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ predictions.

Model / Dataset	σ_8	S_8
ΛCDM (CLASS, Planck 2018 baseline; control)	0.8109	—
SSEE+IS (linear growth, $\gamma_{\rm IS} = 0.5504$, Paper 5)	0.8136	0.8256
SSEE+IS (single-sector ceiling, CLASS, Paper 5)	0.8149	0.827
SSEE, A_s from its own CMB fit (Paper 6)	0.8153	**0.8273**
Planck 2018 [Planck Collaboration, 2020b]	0.812 ± 0.008	0.832 ± 0.013
KiDS-Legacy (raw $\xi_\pm$, Paper 6)	—	0.8265 ± 0.0176
KiDS-1000 [Asgari et al., 2021]	—	0.759 ± 0.024
DES Y3 [DES Collaboration, 2022]	—	0.773 ± 0.025

5.8 $f\sigma_8(z)$ comparison with RSD surveys

The growth-index prediction $\gamma_{\rm IS}$ is directly testable through redshift-space distortions. We compute $f\sigma_8(z) = \Omega_m(z)^\gamma \times \sigma_8 \times D(z)/D(0)$ using the proper integral $D(z)/D(0) = \exp\left[-\int_0^z \Omega_m(z')^\gamma/(1+z')\,dz'\right]$ and compare against published RSD measurements from 6dFGS [Beutler et al., 2012], BOSS DR12 [Alam et al., 2017], eBOSS DR16 [Alam et al., 2021, de Mattia et al., 2021], and DESI DR1 [DESI Collaboration, 2025].

Table 10 shows that SSEE+IS achieves $\chi^2/N = 0.766$ across the six canonical RSD points in the CMB observational sector ($\Omega_{m,\rm CMB} = 0.308881$), compared to 0.860 for ΛCDM. The mean tension is 0.70σ (SSEE) vs 0.73σ (ΛCDM); no SSEE point deviates by more than 1.75σ. This is the identical dataset and method used in Paper 5; we do not include DESI DR1 full-shape $f\sigma_8$ here because no public per-tracer $f\sigma_8$ release exists for DESI DR2 (DR2 is BAO-only, Abdul-Karim et al., 2025), and we use a single self-consistent RSD compilation. The combination of independent growth-rate probes is reviewed in Moresco et al. [2022].

Table 10: $f\sigma_8(z)$ predictions vs the canonical Paper 5 RSD compilation (six peer-reviewed points), computed with the canonical Paper 5 Israel–Stewart growth ODE: $\Omega_{m,\mathrm{CMB}} = 0.308881$ (ω_m-direct) in both the gravitational background and the Poisson source, with $\gamma_{\mathrm{IS}} = 0.5504$ and $\sigma_8 = 0.8136$ (single-sector). Pull $= (f\sigma_8^{\mathrm{SSEE}} - f\sigma_8^{\mathrm{obs}})/\sigma$.

Survey	z_{eff}	Observed	SSEE	ΛCDM	Pull
6dFGRS (Beutler+2012)	0.067	0.423 ± 0.055	0.436	0.444	$+0.24\sigma$
SDSS MGS (Howlett+2015)	0.150	0.490 ± 0.145	0.449	0.459	-0.28σ
BOSS DR12 (Alam+2017)	0.380	0.497 ± 0.045	0.470	0.476	-0.61σ
BOSS DR12 (Alam+2017)	0.510	0.458 ± 0.038	0.471	0.474	$+0.35\sigma$
BOSS DR12 (Alam+2017)	0.610	0.436 ± 0.034	0.468	0.469	$+0.96\sigma$
eBOSS DR16 (Hou+2021)	1.480	0.462 ± 0.045	0.383	0.376	-1.75σ
χ^2/N				SSEE: 0.766	ΛCDM: 0.860

Sector caveat for $f\sigma_8$

The $\chi^2/N = 0.766$ quoted above uses the single matter density $\Omega_m = 0.308881$ in both the background and the Poisson source, with $\gamma_{\mathrm{IS}} = 0.5504$ and $\sigma_8 = 0.8136$ (A_s at the Planck value), finding a mean $f\sigma_8$ tension of 0.70σ (Paper 5, $\approx$ ΛCDM 0.73σ). Against the *raw* BOSS DR12 multipoles, Paper 6 finds the algebraic background fitting marginally better than Planck's ($\chi^2 = 205.55$ vs 207.58 over 222 points, same freedom, each model with its own $\sum m_\nu$).

5.9 CLASS/hi_class cross-check of Bellini–Sawicki α-functions

The Bellini–Sawicki functions [Bellini and Sawicki, 2014] of the SSEE dark-energy sector are validated independently of CAMB with CLASS [Lesgourgues, 2011] in its HI_CLASS extension (Paper 7).

For the Paper 7 action — a k-essence scalar with $K(X) = c_1 X + c_2 X^2$, minimally coupled and with no potential — the Bellini–Sawicki kineticity is

$$\alpha_K^{\mathrm{SSEE}}(0) = 3\,\Omega_{\mathrm{DE}}\,(5 - 3w_0) = 15.591336, \tag{17}$$

with the dark-energy *density* $\Omega_{\mathrm{DE}} = 1 - \Omega_m = 0.691119$, and the sound speed is $c_s^2 = (M_\nu - T_r)/(5M_\nu + 3T_r) = 0.021284$. An independent HI_CLASS integration reproduces c_s^2 to 0.000% and $\alpha_K(0)$ to 0.011% (Paper 7). The kineticity is not observationally accessible: a factor 100 change moves σ_8 by 0.007%. It must not be confused with $s_K = -3w_0(1+w_0) = 3\,s_{\mathrm{DE}}\,s_m = 0.403302$, the pressure dilution rate built from the saturations $s_{\mathrm{DE}} = 0.839950$ and $s_m = 0.160050$ (numbers of the equation of state, not densities), which enters the Hubble screening of Paper 9.

The remaining Bellini–Sawicki functions are zero by construction: $\alpha_T = \alpha_M = \alpha_B = 0$ (exact), since SSEE is a minimal Horndeski scalar with constant M_{Pl} and no kinetic braiding or gravitational-wave speed excess [Abbott et al., 2017]. Together with Eq. (17) this specifies the full set of Bellini–Sawicki EFT functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$, validated against CLASS independently of CAMB.

Note that the raw SSEE *dynamical* background ($s_m = 0.160050$) fed directly to CLASS gives $\chi_r^2 = 90.3$ against Planck 2018 TT. This is expected: the dynamical-sector fraction is not the CMB matter density. The good fit ($\chi_r^2 \approx 1.04$, §5) is recovered with the ω_m-direct

CMB matter density $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.3089$ (no matter factor; OP-8 dissolved), as implemented with CAMB in §4.

6 Falsifiability Criteria

SSEE makes the following testable predictions, each of which would **falsify** the framework if violated:

1. **Peak positions:** $\ell_1 = 220 \pm 3$, $\ell_2 = 536 \pm 5$, $\ell_3 = 813 \pm 8$. A measurement placing ℓ_1 outside this range by more than 2σ would falsify the ω_m-direct CMB matter density.

2. **CMB χ^2_r:** Must remain < 2 when tested against future CMB datasets [Abazajian et al., 2016, Ade et al., 2019]. A significant degradation would indicate the framework cannot maintain coherence.

3. **BAO consistency:** the BAO geometry is governed by the total matter density $\Omega_m = 0.308881$, not by the saturation $1 + w_0 = 0.160050$; SSEE must continue to fit DESI DR2 BAO within 2σ as new data arrive. The algebraic content being tested is the equation of state: a joint fit requiring w_0 more than 2σ from -0.839950, or w_a from -0.669975, would falsify the algebraic derivation. *(An earlier version of this item placed $\Omega_{m,\mathrm{dyn}} = 0.160050$ in the geometry slot; that is a saturation, not a density, and the reading is withdrawn.)*

4. **Cosmic-shear amplitude:** with the background fixed by the algebra and A_s fixed by the model's own CMB fit, the growth sector has no free cosmological parameter and predicts $S_8 = 0.8273$ (Paper 6). A shear measurement placing S_8 more than 3σ from this value would falsify the growth sector.

5. **Growth index:** the full causal IS solution (Paper 5) gives $\gamma_{\mathrm{IS}} = 0.5504 \pm 0.0003 \approx \gamma_\Lambda$ (the earlier estimates ϕ^{-1} and the background-only value of Paper 1, App. A, are superseded), yielding a single-sector ceiling $S_8 = 0.827$ in the CMB observational sector (Table 9). The RSD comparison (Table 10) shows $\chi^2/N = 0.766$ consistently below ΛCDM (0.860) across the six canonical RSD surveys. A direct measurement of $\gamma < 0.50$ or $\gamma > 0.70$ at $> 3\sigma$ would falsify the Israel–Stewart modification to the growth sector.

7 Conclusions

We have demonstrated that SSEE, with the CMB matter density set *forward* by the SSEE algebra (ω_m-direct, $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$, no matter factor), reproduces the Planck 2018 CMB TT power spectrum without introducing fitted dimensionless parameters. The key results are:

- Acoustic peaks at $\ell = 220$, 536, 813, within $\Delta\ell \leq 5$ of Planck 2018.
- TT: $\chi^2_r = 1.042$ over 1971 multipoles (ΛCDM: 1.043).
- TE/EE: $\chi^2_r = 1.040/1.040$, essentially matching ΛCDM.

- Lensing (PP): $\chi^2_r = 0.720$ vs. 0.757 for ΛCDM — SSEE marginally better (both $\chi^2_r < 1$).
- `plik_lite` **TTTEEE + low-ℓ (ω_m-direct):** $\chi^2_{\rm eff} = 1003.586$ with $\{A_s, \tau\}$ minimized, at $H_0 = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.90σ from the `plik_lite` minimum 67.87 ± 0.10 with SSEE-fixed ω_b, ω_c); ΛCDM gives 1003.596 with six free parameters (its own $\sum m_\nu = 0.06$ eV).
- $\Delta\mathrm{BIC} = -26.03$ ($k = 2$ vs $k = 6$, $N = 669$), decisively favouring SSEE over ΛCDM at equal fit quality ($\Delta\chi^2 = -0.010$).
- **Full off-diagonal `plik` MCMC (canonical):** minimum $\chi^2 = 2768.4$ vs 2771.2 for ΛCDM over $N = 2354$ points, $\Delta\mathrm{BIC} = -33.83$ ($k = 2$ vs $k = 6$); -18.31 under the conservative $k = 4$.
- $\Omega_{m,\mathrm{CMB}} = 0.308881$ within 0.88σ of Planck 2018.
- The sound horizon $r_d = 147.17$ Mpc matches ΛCDM to 0.05%.

Combined with the Paper 2 results (0.24σ on the DESI DR2 w_0w_aCDM posterior, and galaxy-cluster dynamical masses in general relativity at the level of ΛCDM: 2.32σ against 2.41σ over the 46 systems of Zhang et al. [2026] with their IGIMF baryon masses), SSEE is consistent with all three major cosmological probes — BAO, cluster masses, and CMB; the cluster masses do not discriminate it from ΛCDM. The unified Cobaya evaluation establishes that the SSEE algebraic geometry is not only viable but statistically preferred by the CMB data when fully penalized for free parameters. Independently of the CMB sector, the SSEE screening cascade of Papers 9–10 takes the measured SH0ES rate as input and, with the full screening fraction, returns $H_0^{\rm glob} = 73.04 \times (1 - 0.069522) = 67.962142\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, a residual of $+4.2 \times 10^{-6}$ from the pure-number target $3(\varphi + \pi)^2$ [Verde et al., 2019, Di Valentino et al., 2021, Riess et al., 2022]; the propagated uncertainty is ± 0.968.

Data Availability. All analysis scripts (`src/p03_cmb/ssee_paper3_cmb.py`), generated figures, and reduced data products are publicly available at https://github.com/mikealmeida1721/SSEE-Cosmologia. The Planck 2018 TT and TE/EE power spectra are from the Planck Legacy Archive (https://irsa.ipac.caltech.edu/data/Planck/) [Planck Collaboration, 2020b]. CAMB v1.6.6 was used for all theoretical power spectrum calculations (https://camb.info).

Acknowledgements. The author thanks the open-source communities behind CAMB, NumPy, Matplotlib, and `emcee` [Foreman-Mackey et al., 2013].

References

K. N. Abazajian et al. CMB-S4 science book, first edition. *arXiv e-prints*, 2016.

B. P. Abbott et al. GW170817: Observation of gravitational waves from a binary neutron star inspiral. *Phys. Rev. Lett.*, 119:161101, 2017. doi: 10.1103/PhysRevLett.119.161101.

M. Abdul-Karim et al. DESI DR2 results II. *Phys. Rev. D*, 112:083515, 2025.

Peter Ade et al. The simons observatory: Science goals and forecasts. *JCAP*, 2019(02): 056, 2019. doi: 10.1088/1475-7516/2019/02/056.

S. Alam et al. The clustering of galaxies in the completed SDSS-III baryon oscillation spectroscopic survey. *MNRAS*, 470:2617, 2017. doi: 10.1093/mnras/stx721.

S. Alam et al. Completed SDSS-IV extended baryon oscillation spectroscopic survey: Cosmological implications from two decades of spectroscopic surveys at the Apache Point Observatory. *Phys. Rev. D*, 103:083533, 2021. doi: 10.1103/PhysRevD.103.083533.

Mike Edison Almeida Vallejo. SSEE: Structural self-energy expansion — framework and bayesian validation, 2026. Paper 1 & 2, preprint.

Marika Asgari et al. KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics. *Astron. Astrophys.*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

Emilio Bellini and Ignacy Sawicki. Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity. *JCAP*, 07:050, 2014. doi: 10.1088/1475-7516/2014/07/050.

F. Beutler et al. The 6df galaxy survey: baryon acoustic oscillations and the local hubble constant. *MNRAS*, 423:3430, 2012. doi: 10.1111/j.1365-2966.2012.21136.x.

Michel Chevallier and David Polarski. Accelerating universes with scaling dark matter. *Int. J. Mod. Phys. D*, 10:213–224, 2001. doi: 10.1142/S0218271801000822.

A. de Mattia et al. The completed SDSS-IV extended baryon oscillation spectroscopic survey: Measurement of the BAO and growth rate of structure of the emission line galaxy sample from the anisotropic power spectrum. *MNRAS*, 501:5616, 2021. doi: 10.1093/mnras/staa3891.

DES Collaboration. Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing. *Phys. Rev. D*, 105:023520, 2022. doi: 10.1103/PhysRevD.105.023520.

DESI Collaboration. DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations. *J. Cosmol. Astropart. Phys.*, 2025:021, 2025. doi: 10.1088/1475-7516/2025/02/021.

Eleonora Di Valentino et al. In the realm of the Hubble tension — a review of solutions. *Class. Quantum Grav.*, 38:153001, 2021. doi: 10.1088/1361-6382/ac086d.

Scott Dodelson. *Modern Cosmology*. Academic Press, San Diego, 2003.

Daniel J. Eisenstein et al. Detection of baryon acoustic oscillations in the correlation function of a large sample of red galaxies. *ApJ*, 633:560–574, 2005. doi: 10.1086/466512.

Wenjuan Fang, Wayne Hu, and Antony Lewis. Crossing the phantom divide with parameterized post-friedmann dark energy. *Phys. Rev. D*, 78:087303, 2008. doi: 10.1103/PhysRevD.78.087303.

D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman. `emcee`: The MCMC hammer. *PASP*, 125:306, 2013.

Andrew Gelman and Donald B. Rubin. Inference from iterative simulation using multiple sequences. *Statist. Sci.*, 7:457–472, 1992. doi: 10.1214/ss/1177011136.

Cullan Howlett, Antony Lewis, Alex Hall, and Anthony Challinor. Cmb power spectrum parameter degeneracies in the era of precision cosmology. *JCAP*, 2012(04):027, 2012. doi: 10.1088/1475-7516/2012/04/027.

Wayne Hu and Naoshi Sugiyama. Small-scale cosmological perturbations: An analytic approach. *ApJ*, 471:542–570, 1996. doi: 10.1086/177989.

Julien Lesgourgues. The Cosmic Linear Anisotropy Solving System (CLASS) I: Overview. *arXiv e-prints*, 2011.

Antony Lewis, Anthony Challinor, and Anthony Lasenby. Efficient normal mode analysis of the cmb. *ApJ*, 538:473–476, 2000. doi: 10.1086/309179.

Eric V. Linder. Exploring the expansion history of the universe. *Phys. Rev. Lett.*, 90: 091301, 2003. doi: 10.1103/PhysRevLett.90.091301.

M. Moresco et al. Unveiling the universe with emerging cosmological probes. *Living Rev. Relativ.*, 25:6, 2022.

S. Perlmutter et al. Measurements of ω and λ from 42 high-redshift supernovae. *ApJ*, 517:565–586, 1999. doi: 10.1086/307221.

Planck Collaboration. Planck 2018 results VI. cosmological parameters. *A&A*, 641:A6, 2020a.

Planck Collaboration. Planck 2018 results. v. cmb power spectra and likelihoods. *A&A*, 641:A5, 2020b. doi: 10.1051/0004-6361/201936386.

Planck Collaboration. Planck 2018 results. XIV. dark energy and modified gravity. *A&A*, 641:A14, 2020c. doi: 10.1051/0004-6361/201833887.

Planck Collaboration. Planck 2018 results. VIII. gravitational lensing. *A&A*, 641:A8, 2020d. doi: 10.1051/0004-6361/201833887.

Planck Collaboration. Planck 2018 results. V. CMB power spectra and likelihoods. *Astron. Astrophys.*, 641:A5, 2020e. doi: 10.1051/0004-6361/201936386.

Adam G. Riess et al. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *AJ*, 116:1009–1038, 1998. doi: 10.1086/300499.

Adam G. Riess et al. A comprehensive measurement of the local value of the Hubble constant with 1 km/s/mpc uncertainty from the Hubble Space Telescope and the SH0ES team. *ApJ*, 934:L7, 2022. doi: 10.3847/2041-8213/ac5c5b.

Uroš Seljak and Matias Zaldarriaga. A line-of-sight integration approach to cosmic microwave background anisotropies. *ApJ*, 469:437–444, 1996. doi: 10.1086/177793.

Joseph Silk. Cosmic black-body radiation and galaxy formation. *ApJ*, 151:459–471, 1968. doi: 10.1086/149449.

Jesús Torrado and Antony Lewis. Cobaya: Code for Bayesian Analysis of hierarchical physical models. *J. Cosmology Astropart. Phys.*, 05:057, 2021. doi: 10.1088/1475-7516/2021/05/057.

Licia Verde, Tommaso Treu, and Adam G. Riess. Tensions between the early and late universe. *Nat. Astron.*, 3:891–895, 2019. doi: 10.1038/s41550-019-0902-0.

Steven Weinberg. The cosmological constant problem. *Rev. Mod. Phys.*, 61:1–23, 1989. doi: 10.1103/RevModPhys.61.1.

D. Zhang, A. H. Zonoozi, and P. Kroupa. Revisiting the missing mass problem in MOND for nearby galaxy clusters. *Phys. Rev. D*, 113:043027, 2026.